BrightPath Tutors Capstone - Final Report

Student: Grace Iliya Rishama

Training account / team: BaseStack Academy Cohort 2

AWS Region: `eu-north-1` (Europe Stockholm)

Date: November 5, 2026

1. What I built

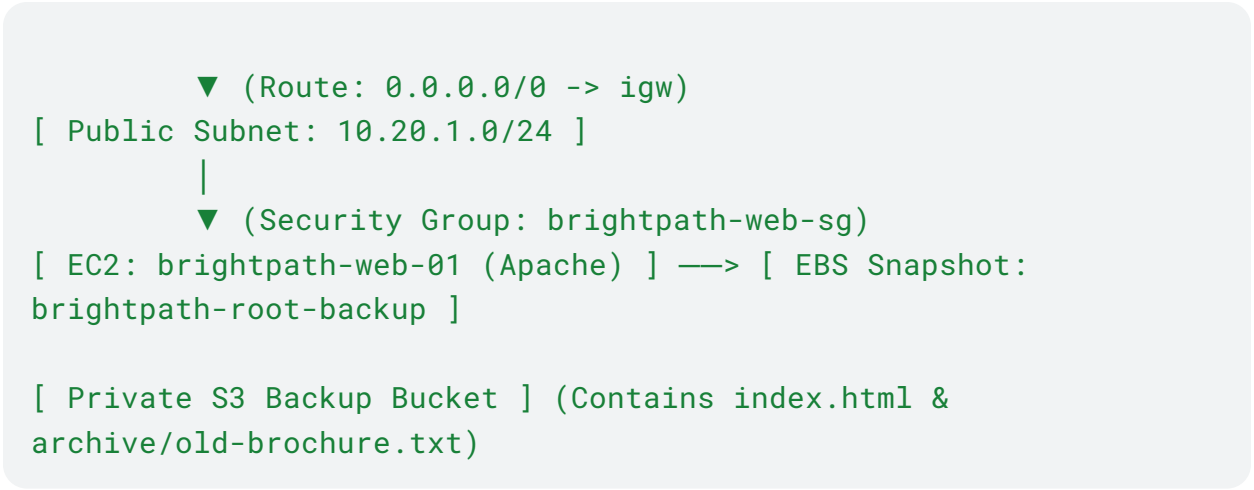
I deployed a public, static website for BrightPath Tutors, an educational business, displaying its tutoring services, opening hours, and contact details. The infrastructure was built entirely via the AWS Management Console without CLI or terminal access. The AWS resources provisioned include:

- A custom VPC (`brightpath-vpc`, CIDR `10.20.0.0/16`) with one public subnet (`brightpath-public-subnet`, CIDR `10.20.1.0/24`).
- An Internet Gateway (`brightpath-igw`) attached to the VPC and connected via a custom route table (`brightpath-public-rt`).
- A custom Web Security Group (`brightpath-web-sg`).
- One Amazon EC2 instance (`brightpath-web-01`, `t3.micro`, Amazon Linux 2023) running an Apache web server initialized automatically via EC2 User Data.
- A private Amazon S3 backup bucket (`brightpath-backup-gir-4821`) with SSE-S3 default encryption, versioning, and an automated lifecycle transition rule.
- An Amazon EBS snapshot (`brightpath-root-backup`) capturing a point-in-time backup of the server's root volume.

2. Architecture

Plaintext

```
None
[ Browser / Visitor ]
         |
         ▼ (HTTP Port 80)
[ Internet Gateway: brightpath-igw ]
         |
```



Traffic Path:

A visitor enters the public IPv4 address into their web browser using the HTTP protocol. The incoming HTTP request travels across the internet and enters the AWS VPC through the attached Internet Gateway (`brightpath-igw`). The custom route table routes the external traffic into the public subnet (`brightpath-public-subnet`), where it is evaluated by the instance firewall (`brightpath-web-sg`). Because port 80 is authorized, the security group permits the packets to reach the Amazon EC2 instance (`brightpath-web-01`), where the Apache HTTP server processes the request and serves the website back along the same path.

3. Security decisions

- **How is HTTP traffic allowed?** Inbound HTTP web traffic is permitted by an inbound rule in the security group `brightpath-web-sg` set to protocol TCP, port 80, and source `0.0.0.0/0` (Anywhere-IPv4).
- **Why did you not open SSH to the internet?** In accordance with the principle of least privilege, opening port 22 (SSH) creates an unnecessary attack vector for brute-force attacks and unauthorized remote shell access; the website is purely static and was deployed fully hands-off via User Data, eliminating any operational need for remote administration access.
- **How is the S3 backup bucket protected?** The S3 bucket has **Block All Public Access** strictly enabled, ensuring no public bucket policies or ACLs expose files to the internet, and all objects are encrypted at rest using server-side encryption (SSE-S3).
- **What does CloudTrail help you review?** AWS CloudTrail records account activity and API requests into an immutable audit log, allowing administrators to audit which IAM identity performed specific resource modifications, the exact source IP, and the event timestamp (e.g., verifying `RunInstances` or `CreateBucket`).

4. Storage and backup

- **What did you put in S3?** I uploaded the website source file `index.html` into the bucket root, created a folder prefix named `archive/`, and uploaded `old-brochure.txt` inside it.
- **What did versioning show?** S3 Bucket Versioning demonstrated that uploading a modified `index.html` preserves the original copy alongside the new one under the same object key, assigning unique Version IDs to prevent accidental overwrites or permanent file deletions.
- **What is an EBS snapshot for?** An EBS snapshot creates a point-in-time, crash-consistent block-level backup of the EC2 instance's root storage volume, enabling rapid disaster recovery or redeployment if the running instance fails or is terminated.
- **What lifecycle rule did you create?** I created a rule named `archive-to-infrequent-access` scoped to the `archive/` prefix that automatically transitions current object versions to the **S3 Standard-IA** (Infrequent Access) storage class after 30 days to reduce storage costs.

5. Database and scaling choices

- **Database Selection (RDS/Aurora vs. DynamoDB):** If BrightPath Tutors later expands to support student bookings, tutor schedules, and payment processing, a relational database like Amazon RDS or Aurora is ideal to handle relational integrity, complex queries, and ACID transactional guarantees across multiple tables. If they only implement a high-traffic timetable lookup with simple key-based queries, DynamoDB would provide fast, low-latency key-value performance.
- **Caching with ElastiCache:** Amazon ElastiCache (Redis/Memcached) can store frequently requested timetable data in memory to reduce repeated database read queries; however, it cannot serve as the sole data store because RAM is volatile and data would be lost during node restarts.
- **High Availability & Scaling (ALB + ASG):** Deploying an Auto Scaling Group (ASG) combined with an Application Load Balancer (ALB) allows the website to handle traffic surges automatically by launching or terminating instances across multiple Availability Zones, with the ALB distributing incoming traffic evenly across healthy nodes.
- **NAT Gateway:** A NAT Gateway is only needed when instances in a private subnet must download software updates without receiving inbound connections from the internet; it is unnecessary here because our single instance sits directly in a public subnet.
- **Storage Choices (EBS vs. EFS):** Amazon EBS provides a dedicated, low-latency boot disk for a single EC2 instance, whereas Amazon EFS provides a scalable, shared network file system accessible simultaneously by multiple Linux instances across Availability Zones.

6. Problem and fix

- **Issue Encountered:** The public IPv4 address of the EC2 instance timed out and failed to load in the browser upon initial deployment and My EC2 instance failed about four different times, i had to delete all my virtual resources on aws console.
- **What was checked in the console:** I inspected the EC2 Instance Details to confirm its public IPv4 address, verified that the instance was running with 2/2 status checks passed, checked the subnet associations on `brightpath-public-rt` to ensure `0.0.0.0/0` targeted `brightpath-igw`, and checked `brightpath-web-sg` inbound rules.
- **How it was fixed:** The browser was automatically prepending `https://` to the address bar, attempting an SSL handshake on port 443 where no listener existed; explicitly typing `http://<PUBLIC-IP>` using standard HTTP port 80 in an incognito window allowed the BrightPath Tutors website to load immediately.
  I later realized that mySG was not created Wizard 1, i created it and it worked.

7. Cost and cleanup

- **Region:** `eu-north-1` (Europe, Stockholm)
- **Instance Type:** `t3.micro` (2 vCPUs, 1 GiB memory)
- **Pricing Source:** AWS EC2 On-Demand Pricing for `eu-north-1` (~$0.0104 per hour)
- **Running Duration:** 15 hours and 17min of Creating and deleting, active testing and evidence capturing.
- **Resources Deleted Post-Audit:**
  1. Terminated EC2 instance `brightpath-web-01`.
  2. Deleted root EBS snapshot `brightpath-root-backup`.
  3. Emptied all versions and deleted S3 bucket `brightpath-backup-gir-4821`.
  4. Detached and deleted Internet Gateway `brightpath-igw`.
  5. Deleted route table `brightpath-public-rt` and subnet `brightpath-public-subnet`.
  6. Deleted security group `brightpath-web-sg` and custom VPC `brightpath-vpc`.

8. Evidence

The following captured screenshots are attached in `submission/EVIDENCE_CHECKLIST.md`:

1. `01-website-browser-load.png`: Browser window displaying the live BrightPath Tutors page at `http://<PUBLIC-IP>`.
2. `02-vpc-route-table.png`: Route table routes showing `0.0.0.0/0` targeting `brightpath-igw`.

3. `03-security-group-rules.png`: Inbound rules of `brightpath-web-sg` showing HTTP port 80 allowed and port 22/SSH absent.
4. `04-s3-bucket-versioning.png`: S3 console showing two distinct versions of `index.html` under the same key.
5. `05-s3-lifecycle-rule.png`: Management tab displaying the `archive-to-infrequent-access` transition rule.
6. `06-ebs-snapshot-completed.png`: EC2 Snapshots table displaying `brightpath-root-backup` in **Completed** status.
7. `07-cloudtrail-event-history.png`: CloudTrail Event history entry showing details for the `RunInstances` event.

9. What I learned

1. **VPC Networking & Routing Fundamentals:** I can clearly explain how traffic moves through an AWS network—specifically how an Internet Gateway acts as the bridge between the internet and a VPC, and why a public subnet requires an explicit `0.0.0.0/0` entry in its route table pointing to that gateway for servers to be publicly reachable.
2. **Layered Cloud Security & Least Privilege:** I can explain why security groups should strictly open only the application ports needed (HTTP port 80) while disallowing administrative interfaces like SSH, and how leaving S3 Block Public Access enabled ensures sensitive operational backups remain isolated from public web traffic.
3. **Decoupled Architecture & Storage Tiers:** I can distinguish between ephemeral compute/block storage (EBS) and durable object storage (S3), and articulate how automated features like S3 lifecycle rules and EBS point-in-time snapshots protect business continuity while optimizing cloud expenditure.